GED Science Practice Test 2

PART A — QUESTIONS

GED-SCI2-001
Section: Science
Type: mcq
Category: Life Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A student wants to find out whether yeast produces more carbon dioxide when sugar is available. The student places yeast and warm water into two identical bottles. Bottle A receives 5 g of sugar, and Bottle B receives no sugar. A balloon is placed over the opening of each bottle to collect gas. Both bottles are kept at the same temperature for 20 minutes.

Which observation would best support the idea that sugar increases carbon dioxide production by yeast?
A) The balloon on Bottle A inflates more than the balloon on Bottle B.
B) The balloon on Bottle B inflates more than the balloon on Bottle A.
C) Both balloons remain completely flat after 20 minutes.
D) Both bottles contain the same amount of warm water.

GED-SCI2-002
Section: Science
Type: mcq
Category: Physical Science
Skill: Using Numbers and Graphics in Science
Prompt:
A student measures how much force is needed to stretch a spring different distances.

Stretch distance (cm): 1, 2, 3, 4
Force (N): 2, 4, 6, 8

Which statement best describes the relationship shown in the table?
A) Force decreases as stretch distance increases.
B) Force is directly proportional to stretch distance.
C) Force stays the same for all stretch distances.
D) Force increases only after 3 cm of stretch.

GED-SCI2-003
Section: Science
Type: mcq
Category: Earth and Space Science
Skill: Reading for Meaning in Science
Prompt:
Basalt forms when lava cools quickly at Earth's surface. Granite forms when magma cools slowly underground. Slow cooling gives mineral crystals more time to grow, while fast cooling produces smaller crystals.

Which statement is best supported by this information?
A) Granite usually has larger mineral crystals than basalt.

B) Basalt forms only from sediment compacted under pressure.
C) Granite forms when lava cools quickly at Earth's surface.
D) Basalt and granite must always have the same crystal size.

GED-SCI2-004
Section: Science
Type: mcq
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
The pancreas helps regulate blood sugar. After a person eats a meal high in carbohydrates, blood sugar rises. The pancreas releases insulin, a hormone that helps cells take in glucose from the blood. As cells take in glucose, blood sugar decreases toward a normal range.

What role does insulin play in this process?
A) It raises body temperature after eating.
B) It helps move glucose from the blood into cells.
C) It digests carbohydrates in the stomach.
D) It prevents cells from using glucose for energy.

GED-SCI2-005
Section: Science
Type: numeric_entry
Category: Physical Science
Skill: Using Numbers and Graphics in Science
Prompt:
A metal cube has a mass of 96 g and a volume of 12 mL.

What is the density of the cube in g/mL?

Type your answer as a whole number.

GED-SCI2-006
Section: Science
Type: multi_select
Category: Earth and Space Science
Skill: Using Numbers and Graphics in Science
Prompt:
Select TWO answers.
A coastal town records the average number of flooding days per year.

Decade | Average flooding days per year
1980s | 3
1990s | 5
2000s | 8
2010s | 13

Which TWO statements are best supported by the data?
A) The average number of flooding days increased over time.
B) The 2010s had more than four times as many flooding days as the 1980s.
C) Flooding days decreased from the 1990s to the 2000s.

D) The table proves that flooding can no longer be reduced.
E) The 2000s had exactly the same flooding average as the 1990s.

GED-SCI2-007
Section: Science
Type: mcq
Category: Life Science
Skill: Using Numbers and Graphics in Science
Prompt:
A scientist records the amount of dissolved oxygen in a stream and the number of mayfly larvae found in samples.

Dissolved oxygen (mg/L): 3, 5, 7, 9
Mayfly larvae counted: 4, 12, 26, 39

Which conclusion is best supported by the data?
A) Mayfly larvae were more common where dissolved oxygen was higher.
B) Mayfly larvae were most common at the lowest dissolved oxygen level.
C) Dissolved oxygen had no relationship to mayfly larvae count.
D) Mayfly larvae caused all oxygen to disappear from the stream.

GED-SCI2-008
Section: Science
Type: mcq
Category: Physical Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A student compares how quickly two liquids evaporate. The student pours 20 mL of water into one dish and 20 mL of rubbing alcohol into another identical dish. Both dishes are placed on the same counter at the same room temperature.

Which factor is being tested?
A) Type of liquid
B) Starting volume of liquid
C) Room temperature
D) Shape of the dish

GED-SCI2-009
Section: Science
Type: mcq
Category: Earth and Space Science
Skill: Reading for Meaning in Science
Prompt:
A model of the carbon cycle includes these steps: plants take in $CO_2$ during photosynthesis, animals release $CO_2$ during respiration, dead organisms can become buried, and burning fossil fuels releases stored carbon back into the atmosphere.

Which process removes $CO_2$ from the atmosphere in this model?
A) Photosynthesis by plants
B) Respiration by animals

C) Burning fossil fuels
D) Decomposition releasing gases

GED-SCI2-010
Section: Science
Type: dropdown
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
A cell uses mitochondria to release usable energy from food molecules. In this process, mitochondria are mainly involved in [D1], while the cell membrane mainly helps [D2].
Dropdowns:
D1) cellular respiration | protein storage | DNA copying
D2) control what enters and leaves the cell | make sunlight into sugar | store inherited traits

GED-SCI2-011
Section: Science
Type: drag_drop
Category: Physical Science
Skill: Designing and Interpreting Science Experiments
Prompt:
Match each target with the correct tile.
Tiles:
A) Amount of detergent added to each oil sample
B) Time required for oil to break into small droplets
C) Same volume of oil and water in each container
D) Color of the container label
Targets:
T1) Independent variable
T2) Dependent variable
T3) Controlled variable

GED-SCI2-012
Section: Science
Type: mcq
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
A person inhales air into the lungs. Oxygen moves from tiny air sacs in the lungs into nearby blood vessels. At the same time, carbon dioxide moves from the blood into the air sacs and is exhaled.

Which body systems work together most directly in this exchange?
A) Respiratory and circulatory systems
B) Digestive and skeletal systems
C) Nervous and reproductive systems
D) Muscular and integumentary systems

GED-SCI2-013
Section: Science
Type: mcq
Category: Physical Science

Skill: Using Numbers and Graphics in Science
Prompt:
Four liquids are tested to see whether they conduct electricity.

Liquid | Bulb in circuit
Distilled water | Off
Salt water | Bright
Sugar water | Off
Lemon juice | Dim

Which liquid allowed the most electric current to pass through the circuit?
A) Distilled water
B) Salt water
C) Sugar water
D) Lemon juice

GED-SCI2-014
Section: Science
Type: numeric_entry
Category: Earth and Space Science
Skill: Using Numbers and Graphics in Science
Prompt:
A glacier moves 84 meters in 12 months.

What is the glacier's average movement rate in meters per month?

Type your answer as a whole number.

GED-SCI2-015
Section: Science
Type: mcq
Category: Life Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A researcher studies whether a certain soil bacterium helps pea plants grow taller. Group 1 receives normal soil. Group 2 receives the same soil plus the bacterium. Both groups receive the same light, water, pot size, and type of pea seed.

Why is Group 1 included?
A) It provides a comparison group without the added bacterium.
B) It proves that all bacteria are harmful to plants.
C) It makes the pea seeds genetically identical.
D) It prevents plants in Group 2 from growing.

GED-SCI2-016
Section: Science
Type: mcq
Category: Physical Science
Skill: Reading for Meaning in Science
Prompt:

A student notices that a cold glass of water becomes covered with tiny drops of liquid on the outside. Water vapor in the surrounding air touches the cold glass and changes into liquid water.

Which process forms the drops on the outside of the glass?
A) Melting
B) Condensation
C) Freezing
D) Combustion

GED-SCI2-017
Section: Science
Type: dropdown
Category: Physical Science
Skill: Reading for Meaning in Science
Prompt:
When a ball is held above the ground, it has gravitational potential energy. As the ball falls, much of this energy changes into [D1]. When the ball hits the ground, some energy is transferred as [D2].
Dropdowns:
D1) kinetic energy | chemical energy | nuclear energy
D2) sound and thermal energy | stored food energy | magnetic energy

GED-SCI2-018
Section: Science
Type: multi_select
Category: Life Science
Skill: Designing and Interpreting Science Experiments
Prompt:
Select TWO answers.
A class tests how different amounts of sleep affect reaction time. Students record their hours of sleep and then complete the same reaction-time test on a computer.

Which TWO choices would improve the design of this investigation?
A) Use the same reaction-time test for every student.
B) Collect data from a larger number of students.
C) Let students choose different tests based on preference.
D) Record only the fastest student's result.
E) Change the computer program halfway through the study.

GED-SCI2-019
Section: Science
Type: mcq
Category: Earth and Space Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A student wants to test how slope affects soil erosion. The student prepares three trays with the same soil type and soil amount. Each tray is set at a different angle. The same amount of water is poured over each tray for the same amount of time, and the mass of soil washed away is measured.

What is the dependent variable?
A) Angle of the tray
B) Type of soil

C) Mass of soil washed away
D) Amount of water poured

GED-SCI2-020
Section: Science
Type: mcq
Category: Physical Science
Skill: Using Numbers and Graphics in Science
Prompt:
A student records the temperature of a substance as it cools.

Time (min): 0, 2, 4, 6, 8
Temperature (°C): 90, 78, 66, 54, 42

What is the average cooling rate from 0 to 8 minutes?
A) 4°C/min
B) 6°C/min
C) 8°C/min
D) 12°C/min

GED-SCI2-021
Section: Science
Type: mcq
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
In a population of small mammals, some individuals have thicker fur than others. During several unusually cold winters, mammals with thicker fur survive at higher rates and leave more offspring. Over many generations, thick fur becomes more common in the population.

Which process is described?
A) Natural selection
B) Evaporation
C) Electrical conduction
D) Rock formation

GED-SCI2-022
Section: Science
Type: mcq
Category: Physical Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A student claims that dark-colored surfaces absorb more radiant energy than light-colored surfaces. The student places a black paper square and a white paper square under the same lamp for 10 minutes and records the final temperature of each square.

Which result would best support the student's claim?
A) The black paper reaches a higher temperature than the white paper.
B) The white paper reaches a higher temperature than the black paper.
C) Both paper squares remain at the same temperature.
D) The lamp turns off before the test begins.

GED-SCI2-023
Section: Science
Type: dropdown
Category: Earth and Space Science
Skill: Reading for Meaning in Science
Prompt:
In the water cycle, water that soaks into soil may become [D1], while water that flows over land into rivers is called [D2].
Dropdowns:
D1) groundwater | magma | ozone
D2) runoff | orbit | radiation

GED-SCI2-024
Section: Science
Type: mcq
Category: Life Science
Skill: Using Numbers and Graphics in Science
Prompt:
A family history shows that a certain genetic condition appears only when a child inherits two recessive alleles. Two parents are both carriers, meaning each parent has one dominant allele and one recessive allele.

For each child, what is the probability of inheriting two recessive alleles?
A) 0%
B) 25%
C) 50%
D) 100%

GED-SCI2-025
Section: Science
Type: multi_select
Category: Physical Science
Skill: Designing and Interpreting Science Experiments
Prompt:
Select TWO answers.
A student investigates how blade length affects the power produced by a small model wind turbine. The student tests three blade lengths in front of the same fan and measures voltage.

Which TWO factors should be kept constant?
A) Distance between the fan and turbine
B) Fan speed
C) Blade length
D) Voltage produced
E) Final data values

GED-SCI2-026
Section: Science
Type: mcq
Category: Life Science
Skill: Designing and Interpreting Science Experiments
Prompt:

A scientist is testing whether a new disinfectant reduces the number of microbes on a surface. Ten identical tiles are wiped with plain water, and ten identical tiles are wiped with the disinfectant. After 30 minutes, samples from all tiles are placed on growth plates and colonies are counted.

Which evidence would best support the claim that the disinfectant works?
A) Tiles wiped with disinfectant have fewer colonies on average.
B) Tiles wiped with plain water have fewer colonies on average.
C) All tiles grow the exact same number of colonies.
D) Colonies are counted before the tiles are wiped.

GED-SCI2-027
Section: Science
Type: mcq
Category: Physical Science
Skill: Reading for Meaning in Science
Prompt:
A chemical hand warmer contains substances that react when exposed to air. The reaction releases thermal energy, making the packet feel warm.

Which statement best describes the reaction?
A) It is exothermic because it releases thermal energy.
B) It is endothermic because it releases thermal energy.
C) It destroys energy and creates cold.
D) It changes thermal energy into mass.

GED-SCI2-028
Section: Science
Type: mcq
Category: Physical Science
Skill: Designing and Interpreting Science Experiments
Prompt:
A student wants to compare how well different materials reduce sound. The student places each material between a speaker and a sound meter. The speaker volume and distance from the sound meter stay the same for every trial.

Why should the speaker volume stay the same?
A) So the type of material is the only main variable being tested
B) So every material produces the same sound
C) So the sound meter stops measuring decibels
D) So sound waves no longer travel through air

GED-SCI2-029
Section: Science
Type: mcq
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
A pond ecosystem contains algae, small insects, fish, and herons. Algae use sunlight to make food. Small insects eat algae, fish eat insects, and herons eat fish.

If algae greatly decrease, which effect is most likely to happen first?
A) Small insects would have less food available.
B) Herons would immediately begin photosynthesis.
C) Fish would stop needing energy.
D) The pond would gain unlimited producers.

GED-SCI2-030
Section: Science
Type: mcq
Category: Physical Science
Skill: Using Numbers and Graphics in Science
Prompt:
A student mixes 40 g of solution X with 25 g of solution Y in an open beaker. After a reaction, some gas escapes, and the remaining liquid has a mass of 58 g.

How much mass escaped as gas?
A) 3 g
B) 7 g
C) 15 g
D) 65 g

GED-SCI2-031
Section: Science
Type: mcq
Category: Life Science
Skill: Using Numbers and Graphics in Science
Prompt:
A nutrition label shows that one serving of a snack contains 180 calories and 6 g of protein. A person eats 3 servings.

How many grams of protein does the person eat?
A) 6 g
B) 12 g
C) 18 g
D) 540 g

GED-SCI2-032
Section: Science
Type: numeric_entry
Category: Physical Science
Skill: Using Numbers and Graphics in Science
Prompt:
A machine transfers 240 kJ of useful energy from 600 kJ of energy input.

What percent of the input energy is useful energy?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI2-033
Section: Science
Type: mcq

Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
Antibiotics kill many bacteria, but a few bacteria may have traits that allow them to survive. If antibiotics are used often, the surviving bacteria can reproduce and pass on those traits.

What is the best explanation for why antibiotic-resistant bacteria may become more common?
A) Resistant bacteria are more likely to survive and reproduce after antibiotic exposure.
B) Antibiotics cause every bacterium to become harmless.
C) Bacteria stop reproducing whenever antibiotics are present.
D) Nonresistant bacteria choose to become resistant.

GED-SCI2-034
Section: Science
Type: mcq
Category: Life Science
Skill: Reading for Meaning in Science
Prompt:
Vaccines expose the immune system to a harmless form or part of a pathogen. This helps the body produce memory cells that can respond faster if the real pathogen enters the body later.

Which statement best describes the purpose of memory cells?
A) They store oxygen for muscles.
B) They help the immune system respond faster during future exposure.
C) They digest food into smaller molecules.
D) They prevent the body from making antibodies.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI2-001 — Correct: A — Correct Answer: The balloon on Bottle A inflates more than the balloon on Bottle B. — Explanation: More balloon inflation indicates more gas production, supporting the claim that sugar increases carbon dioxide production by yeast.

GED-SCI2-002 — Correct: B — Correct Answer: Force is directly proportional to stretch distance. — Explanation: The force doubles when the stretch distance doubles, showing a direct proportional relationship.

GED-SCI2-003 — Correct: A — Correct Answer: Granite usually has larger mineral crystals than basalt. — Explanation: Granite cools slowly underground, giving crystals more time to grow.

GED-SCI2-004 — Correct: B — Correct Answer: It helps move glucose from the blood into cells. — Explanation: Insulin lowers blood sugar by helping cells take in glucose.

GED-SCI2-005 — Correct: 8 — Tolerance: exact — Correct Answer: 8 — Explanation: Density = mass ÷ volume = 96 g ÷ 12 mL = 8 g/mL.

GED-SCI2-006 — Correct: A,B — Correct Answer: The average number of flooding days increased over time. | The 2010s had more than four times as many flooding days as the 1980s. — Explanation: Flooding days increased from 3 to 13, and 13 is more than four times 3.

GED-SCI2-007 — Correct: A — Correct Answer: Mayfly larvae were more common where dissolved oxygen was higher. — Explanation: The number of mayfly larvae increased as dissolved oxygen increased.

GED-SCI2-008 — Correct: A — Correct Answer: Type of liquid — Explanation: The type of liquid is the factor intentionally changed between the two dishes.

GED-SCI2-009 — Correct: A — Correct Answer: Photosynthesis by plants — Explanation: During photosynthesis, plants take in $CO_2$ from the atmosphere.

GED-SCI2-010 — Correct: D1=cellular respiration; D2=control what enters and leaves the cell — Correct Answer: D1=cellular respiration; D2=control what enters and leaves the cell — Explanation: Mitochondria release usable energy during cellular respiration, while the cell membrane regulates movement into and out of the cell.

GED-SCI2-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Amount of detergent added to each oil sample; T2=Time required for oil to break into small droplets; T3=Same volume of oil and water in each container — Explanation: The amount of detergent is changed, the time is measured, and the oil and water volumes are controlled.

GED-SCI2-012 — Correct: A — Correct Answer: Respiratory and circulatory systems — Explanation: The respiratory system exchanges gases in the lungs, and the circulatory system transports those gases in the blood.

GED-SCI2-013 — Correct: B — Correct Answer: Salt water — Explanation: The bright bulb shows that salt water allowed the most current to pass through the circuit.

GED-SCI2-014 — Correct: 7 — Tolerance: exact — Correct Answer: 7 — Explanation: Average rate = 84 meters ÷ 12 months = 7 meters per month.

GED-SCI2-015 — Correct: A — Correct Answer: It provides a comparison group without the added bacterium. — Explanation: Group 1 is the control group, allowing comparison with the bacterium-treated group.

GED-SCI2-016 — Correct: B — Correct Answer: Condensation — Explanation: Condensation occurs when water vapor changes into liquid water.

GED-SCI2-017 — Correct: D1=kinetic energy; D2=sound and thermal energy — Correct Answer: D1=kinetic energy; D2=sound and thermal energy — Explanation: As the ball falls, potential energy changes into kinetic energy, and impact transfers some energy as sound and heat.

GED-SCI2-018 — Correct: A,B — Correct Answer: Use the same reaction-time test for every student. | Collect data from a larger number of students. — Explanation: Using the same test controls the method, and a larger sample improves reliability.

GED-SCI2-019 — Correct: C — Correct Answer: Mass of soil washed away — Explanation: The dependent variable is what is measured in response to changing the tray angle.

GED-SCI2-020 — Correct: B — Correct Answer: 6°C/min — Explanation: Temperature dropped from 90°C to 42°C, a 48°C decrease over 8 minutes. The rate is 48 ÷ 8 = 6°C/min.

GED-SCI2-021 — Correct: A — Correct Answer: Natural selection — Explanation: Thick fur became more common because mammals with that trait survived and reproduced more successfully in cold conditions.

GED-SCI2-022 — Correct: A — Correct Answer: The black paper reaches a higher temperature than the white paper. — Explanation: If dark surfaces absorb more radiant energy, the black paper should warm more than the white paper.

GED-SCI2-023 — Correct: D1=groundwater; D2=runoff — Correct Answer: D1=groundwater; D2=runoff — Explanation: Water that soaks into soil can become groundwater, while water flowing over land is runoff.

GED-SCI2-024 — Correct: B — Correct Answer: 25% — Explanation: Two carrier parents can produce offspring with two recessive alleles in 1 out of 4 possible allele combinations.

GED-SCI2-025 — Correct: A,B — Correct Answer: Distance between the fan and turbine | Fan speed — Explanation: These factors should stay constant so blade length is the main variable being tested.

GED-SCI2-026 — Correct: A — Correct Answer: Tiles wiped with disinfectant have fewer colonies on average. — Explanation: Fewer colonies after disinfectant treatment would support the claim that the disinfectant reduces microbes.

GED-SCI2-027 — Correct: A — Correct Answer: It is exothermic because it releases thermal energy. — Explanation: An exothermic reaction releases thermal energy to its surroundings.

GED-SCI2-028 — Correct: A — Correct Answer: So the type of material is the only main variable being tested — Explanation: Keeping volume constant helps ensure that differences in sound level are due to the material.

GED-SCI2-029 — Correct: A — Correct Answer: Small insects would have less food available. — Explanation: Since small insects eat algae, a decrease in algae would first reduce their food supply.

GED-SCI2-030 — Correct: B — Correct Answer: 7 g — Explanation: The starting mass was 40 g + 25 g = 65 g. The remaining mass was 58 g, so 7 g escaped.

GED-SCI2-031 — Correct: C — Correct Answer: 18 g — Explanation: Three servings contain 3 × 6 g = 18 g of protein.

GED-SCI2-032 — Correct: 40 — Tolerance: exact — Correct Answer: 40 — Explanation: 240 ÷ 600 × 100 = 40%, so 40% of the input energy is useful energy.

GED-SCI2-033 — Correct: A — Correct Answer: Resistant bacteria are more likely to survive and reproduce after antibiotic exposure. — Explanation: Antibiotics create conditions where resistant bacteria have a survival advantage and can pass on resistance traits.

GED-SCI2-034 — Correct: B — Correct Answer: They help the immune system respond faster during future exposure. — Explanation: Memory cells allow the immune system to recognize and respond more quickly to a pathogen later.